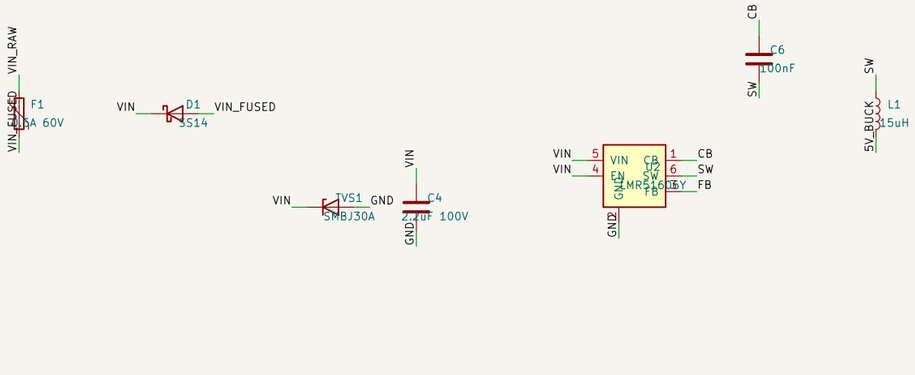
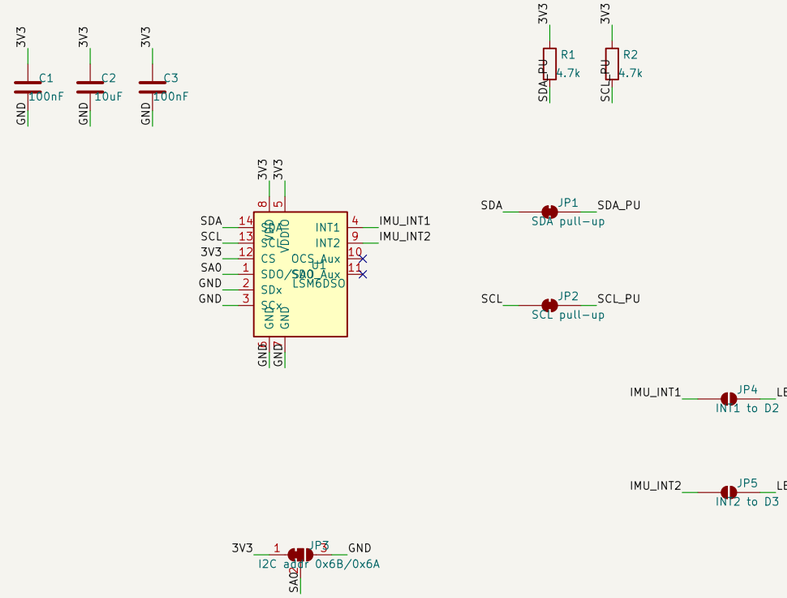
Power: NMEA 2000 NET-S or a 5–30 V lead on the J4 pads -> fuse -> reverse diode -> TVS -> LMR51606 buck 5 V -> diode-OR into the XIAO 5V pin



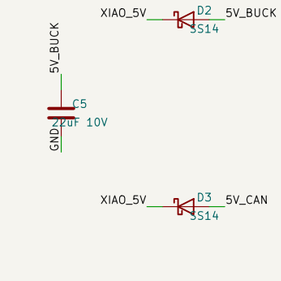
Test pads (rev B): one per layer-crossing net, bottom side, 2.54 mm grid



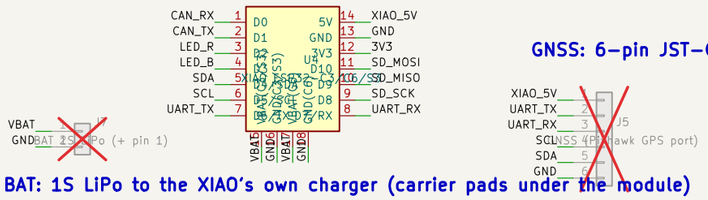
IMU: LSM6DSO on I2C, address 0x6B by default (JP3), as on the Takuma node



Controller: Seeed XIAO ESP32-C3 / C6 / S3 (pads 1–14 are common to all three)

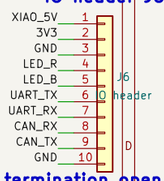


BAT: 1S LiPo to the XIAO's own charger (carrier pads under the module)

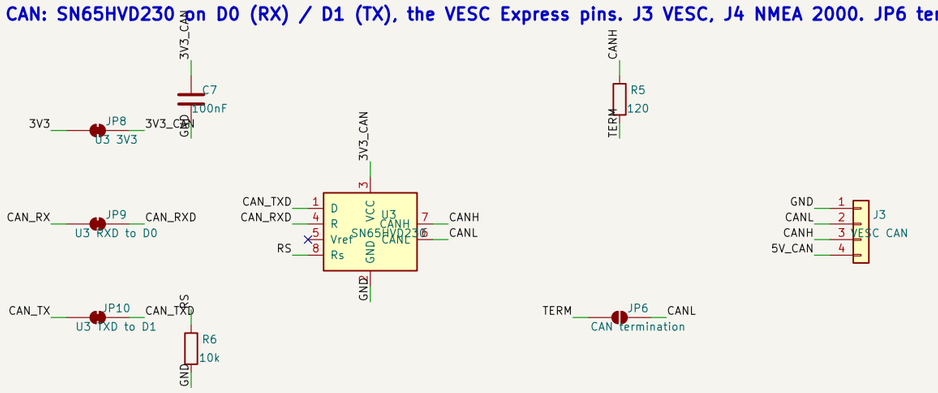


GNSS: 6-pin JST-GH, Pixhawk GPS pinout, UART on D6/D7 + the

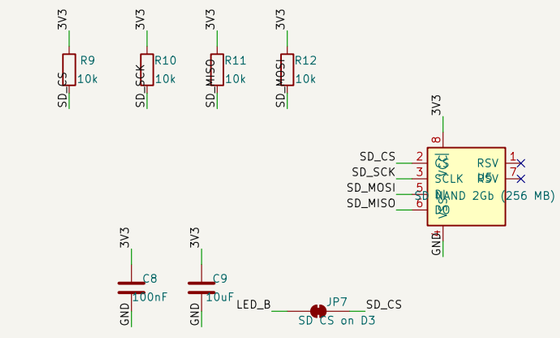
IO header J6: 10



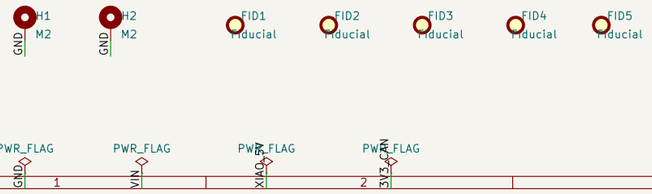
CAN: SN65HVD230 on D0 (RX) / D1 (TX), the VESC Express pins. J3 VESC, J4 NMEA 2000. JP6 termination open. JP



Log storage: SD NAND on SPI, under the XIAO



Mechanical: M2 holes at the bow end



XIAO footprint derived from Seeed OPL_Kicad_Library, CC BY-SA 4.0
Circuit origin: the Takuma project XIAO node (XIAO ESP32-C3 + LSM6DSO + SN65HVD230)
Generated from wavehat_design.py by gen_schematic.py — edit the design file, not this one

Sheet: /
File: wavehat.kicad_sch

Title: WaveHat: sea-state sensor and CAN node for the Seeed XIAO

Size: A3	Date: 2026-11-07	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 1/1